runs, we expect to find opposite spins (1⁄3)(1) + (2⁄3)(1⁄4) = 1⁄2 of the time, as found by the data.

You may find it odd that the assumption of locality yields a higher spin correlation (greater than 50 percent) than what we find with standard quantum mechanics (exactly 50 percent); the long-range entanglement of quantum mechanics, you'd think, should yield a greater correlation. In fact, it does. A way to think about it is this: With only a 50 percent correlation over all measurements, quantum mechanics yields 100 percent correlation for measurements in which the left and right detector axes are chosen to point in the same direction. In the local universe of Einstein, Podolsky, and Rosen, a greater than 55 percent correlation over all measurements is required to ensure 100 percent agreement when the same axes are chosen. Roughly, then, in a local universe, a 50 percent correlation over all measurements would entail *less* than a 100 percent correlation when the same axes are chosen—i.e., less of a correlation than what we find in our nonlocal quantum universe.

19. You might think that an instantaneous collapse would, from the get-go, fall afoul of the speed limit set by light and therefore ensure a conflict with special relativity. And if probability waves were indeed like water waves, you'd have an irrefutable point. That the value of a probability wave suddenly dropped to zero over a huge expanse would be far more shocking than all of the water in the Pacific Ocean's instantaneously becoming perfectly flat and ceasing to move. But, quantum mechanics practitioners argue, probability waves are *not* like water waves. A probability wave, although it describes matter, is not a material thing itself. And, such practitioners continue, the speed-of-light barrier applies only to material objects, things whose motion can be directly seen, felt, detected. If an electron's probability wave has dropped to zero in the Andromeda galaxy, an Andromedan physicist will merely fail, with 100 percent certainty, to detect the electron. Nothing in the Andromedan's observations reveals the sudden change in the probability wave associated with the successful detection, say, of the electron in New York City. As long as the electron itself does not travel from one place to another at greater than light speed, there is no conflict with special relativity. And, as you can see, all that has happened is that the electron was found to be in New York City and not anywhere else. Its speed never even entered the discussion. So, while the instantaneous collapse of probability is a framework that comes with puzzles and problems (discussed more fully in Chapter 7), it need not necessarily imply a conflict with special relativity.